C O U R S E - E N D P R O J E C T R E P O R T

O n

Exploratory Data Analysis and Visualization of News Article Popularity Analysis

Submitted to fulfill the requirements for the Award of CIE Marks (10) for the B.E.(IT), III-SEM,
Couse: **Exploratory Data Analysis & Visualization - EDAV (22ADC31N)**

By

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CERTIFICATE

This is to certify that the course-end project report entitled **"Exploratory Data Analysis and Visualization of News Article Popularity Analysis"** is a bonafide record of the work carried out by **Avikshith Reddy Sarasam(1601-25-737-071)**, in partial fulfilment of the requirements of the course **Exploratory Data Analysis and Visualization (22ADC31N)** of the programme **B.E(IT), III-SEM**, during the academic year **2026–2027**.

The project was carried out under my guidance and supervision. To the best of my knowledge, the work reported here is original and has not been submitted to any other institution or university for the award of any degree, diploma or certificate.

Place: Hyderabad

Date: ___/___/____

Course Faculty

(Asst Prof G.Srikanth)

DECLARATION

I / We hereby declare that the project report entitled **"Exploratory Data Analysis and Geospatial Visualization of Youth Hostel Availability Across Indian States"** submitted to **Dept of IT, CBIT** is a record of original work done by me under the guidance of **G.Srikanth , Asst.Professor, IT, CBIT**.

The data used was taken from the dataset supplied for the course; all external sources, tools and libraries have been acknowledged in the References. This work has not been submitted, in part or in full, for any other course, degree or diploma.

Place:

Date:

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[Name and Roll No]

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Abstract

The rapid adoption of Electric Vehicles (EVs) is a key indicator of the global transition toward sustainable transportation and clean energy mobility. Understanding the factors associated with electric vehicle adoption and usage helps identify patterns in consumer preference across propulsion architectures, vehicle manufacturers, model years, geographic counties, and Clean Alternative Fuel Vehicle (CAFV) policy tiers. This project uses the real-world Electric Vehicle Population dataset (Washington State DOL, 2024), containing 177,866 registered vehicles and 17 primary attributes—alongside a high-dimensional feature-engineered telemetry and one-hot encoded matrix spanning 10,240 columns—to examine the distribution of certified electric range and the relationships between vehicle characteristics and regional adoption.

The project follows an end-to-end data analysis workflow using Python, incorporating data quality assessment, data cleaning, descriptive statistics, exploratory data analysis (EDA), correlation analysis, time-series resampling, and visualization across the five Course Outcomes (CO1 –CO5). The analysis investigates variations in adoption and electric range across Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs), manufacturers, model years, and utility territories. NumPy (Harris et al., 2020) and Pandas (McKinney, 2022) are used for numerical computation and data manipulation, while Matplotlib (Hunter, 2007) and Seaborn (Waskom, 2021) support graphical analysis.

The results indicate that fleet composition is strongly dominated by BEVs at 78.52% (139,662 vehicles), compared with 21 .48% (38,204 vehicles) for PHEVs. Among vehicles with certified non-zero EPA range records, electric range is bimodally distributed, with a median of 215.0 miles (mean 208.45 miles) for BEVs and a median of 27.0 miles (mean 31 .84 miles) for PHEVs. Market adoption is highly concentrated among leading manufacturers, with Tesla accounting for 44.78% (79,659 vehicles) of all registrations, and King County representing 52.21 % (92,860 vehicles) of state-wide volume. Overall, this project demonstrates how exploratory data analysis and visualization can provide actionable insights into electric vehicle adoption and usage patterns.

Keywords: Electric Vehicle Adoption, Exploratory Data Analysis, Data Visualization, Python, Battery Electric Vehicles, Correlation Analysis.

1. Introduction

The rapid growth of electrified transportation has transformed how households commute, how automotive manufacturers design powertrains, and how electric utilities manage regional grid demand. Modern electric vehicles operate under two primary propulsion architectures: Battery Electric Vehicles (BEVs), which rely solely on high-voltage traction batteries and electric motors, and Plug-in Hybrid Electric Vehicles (PHEVs), which pair a rechargeable battery pack with an internal combustion engine. In this environment, vehicle registration volume and certified all-electric range serve as fundamental measures of technological maturity and consumer adoption.

However, electric vehicle adoption and usage characteristics vary considerably across manufacturers, model years, and geographical regions. Some vehicle makes attract massive consumer volume through long-range BEV platforms exceeding 250 miles per charge, while others focus on short-range commuter PHEVs providing 20 to 40 miles of electric range. Furthermore, statutory tax incentives under Clean Alternative Fuel Vehicle (CAFV) policies depend directly on battery range thresholds, creating distinct clusters in registration behavior.

This project uses the real-world Electric Vehicle Population dataset (Washington State DOL, 2024), which contains 177,866 registered vehicles and 17 original attributes, supplemented by a 10,240-column sparse feature expansion matrix encoding VIN prefix patterns, postal census tracts, utility territories, and charging load sub-channels. The dataset provides detailed information about vehicle make, model, model year, propulsion type, CAFV eligibility, certified electric range, base MSRP, legislative district, electric utility provider, and exact geographic coordinates.

The project follows a systematic exploratory data analysis workflow using Python organized across Course Outcomes CO1 through CO5. NumPy is used for multi-dimensional array computation, while Pandas supports data cleaning, missing-value imputation, hierarchical indexing, relational joins, descriptive statistics, and time-series aggregation. Matplotlib and Seaborn are used to visualize distributions, compare manufacturers, examine model-year progression, and explore statistical relationships between vehicle characteristics.

2. Statement of the problem and objectives

Problem statement: To analyze and visualize the adoption and usage patterns ofelectric vehicles by examining registration counts and certified electric range across propulsion types (BEV vs. PHEV), manufacturers, model years, and geographical counties, and to identify statistical patterns related to battery range progression and clean-fuel policy eligibility.

2.1.Objective

1. Examine the Electric Vehicle Population dataset (177,866 rows) and evaluate its quality, completeness, consistency, and high-dimensional matrix scaling (10,240 columns).
2. Clean and prepare the data for analysis: standardize column names, handle missing geographic entries, and impute unrecorded zero-mile EPA range entries.
3. Perform Exploratory Data Analysis (EDA) on vehicle registrations, BEV versus PHEV range distributions, manufacturer market shares, and model-year growth trends.
4. Investigate the relationship between certified electric range and vehicle features such as propulsion architecture, model year, base MSRP, and CAFV eligibility.
5. Conduct statistical and correlation analysis: apply the Mann-Whitney U test and Spearman rank correlation to quantify differences and associations.
6. Evaluate geospatial concentration across counties and electric utilities to support charging infrastructure planning.
7. Present evidence-based findings, recommendations, and conclusions on electric vehicle adoption, mindful ofdataset limitations.

2.2.Questions of the Research

- How is certified electric range distributed across the fleet, and how does typical range differ between BEVs and PHEVs?
- Which automotive manufacturers and vehicle models capture the highest and lowest market shares?
- How has electric vehicle adoption changed across vehicle model years and quarterly registration periods?
- How does certified electric range relate to vehicle parameters such as model year, propulsion type, base MSRP, and CAFV tax eligibility?
- How concentrated are electric vehicle registrations across urban versus rural counties and electric utility service areas?

3. Dataset Description

The analysis uses the Electric Vehicle Population dataset published by the Washington State Department of Licensing (Washington State DOL, 2024) via Kaggle and Data.gov. Each row represents an individual registered Battery Electric Vehicle (BEV) or Plug-in Hybrid Electric Vehicle (PHEV). The dataset contains 177,866 records and 17 base columns, accompanied by a 10,240-column feature-engineered matrix (`X_wide_ev`) constructed to evaluate high-dimensional numerical array operations under Course Outcome 1.

The primary measures of adoption and capability are vehicle registration counts and EPA-certified electric range (represented by the `Electric Range` variable in miles). Other attributes describe the vehicle manufacturer (`Make`), `Model`, `Model Year`, `Electric Vehicle Type`, `Clean Alternative Fuel Vehicle (CAFV) Eligibility`, `Base MSRP`, `County`, `City`, `Postal Code`, `Legislative District`, `Electric Utility`, and WKT spatial coordinates (`Vehicle Location`).

Table 1: Columns and attributes of the dataset

S. No.	Column Name	Data Type	Non-Null Count	Unique Values
1	VIN (1-10)	str (object)	177,866	10,830
2	County	str (object)	177,861	196
3	City	str (object)	177,861	723
4	State	str (object)	177,866	46
5	Postal Code	float64	177,861	862
6	Model Year	int64	177,866	22
7	Make	str (object)	177,866	40
8	Model	str (object)	177,866	139
9	Electric Vehicle Type	str (object)	177,866	2
10	Clean Alternative Fuel Vehicle (CAFV) Eligibility	str (object)	177,866	3

S. No.	Column Name	Data Type	Non-Null Count	Unique Values

11	Electric Range	float64	177,866	102
12	Base MSRP	float64	177,866	31
13	Legislative District	float64	177,477	49
14	DOL Vehicle ID	int64	177,866	177,866
15	Vehicle Location	str (object)	177,857	861
16	Electric Utility	str (object)	177,861	76
17	2020 Census Tract	float64	177,861	2,115
18	X_wide_ev (Engineered Telemetry & One-Hot Matrix)	float32	1,000 (sample)	10,240 columns

4. Methodology and Tools

The project follows a systematic data analysis workflow to study patterns related to electric vehicle adoption and usage. The methodology consists of data loading and inspection, data quality assessment, data cleaning and preparation, exploratory data analysis (EDA), statistical testing, correlation analysis, time-series resampling, and visualization. Python libraries are used to implement each step to ensure consistency and reproducibility.

The workflow starts by loading the Electric Vehicle Population dataset into a Pandas DataFrame. Data quality checks identify missing values, duplicate records, and unrecorded `0`-mile range entries. Exploratory Data Analysis is performed to examine the distribution of certified electric range and compare adoption across vehicle categories (`BEV` vs. `PHEV`), manufacturers, and model years. Differences in electric range between BEV and PHEV architectures are evaluated using the Mann-Whitney U test (Virtanen et al., 2020), and Spearman rank correlation quantifies relationships across numerical variables.

Table 2: Tools Used

Purpose	**Library / Tool**
Programming language	Python
Data handling and manipulation	Pandas
Numerical operations	NumPy
Static data visualization	Matplotlib, Seaborn
Interactive charts and visualizations	Plotly
Statistical analysis	SciPy
Development environment	Google Colab and Jupyter Notebook
Dataset storage and access	CSV files

5. Assessing and Cleaning Data Quality

Data quality assessment is an essential step of exploratory data analysis because the reliability of findings depends on the completeness and consistency ofthe input data. Prior to statistical analysis and visualization, the Electric Vehicle Population dataset was inspected to understand its structure, missing observations, and value consistency.

5.1.Inspection of Data Set

The data was loaded into a Pandas DataFrame from CSV format. The dataset comprises 177,866 vehicles and 17 base features, supplemented by the 10,240-column wide feature matrix (`X_wide_ev`). Column headers were stripped of whitespace, and data types were verified to distinguish numerical attributes (`Model Year`, `Electric Range`, `Base MSRP`, `Legislative District`) from categorical descriptors (`Make`, `Model`, `Electric Vehicle Type`, `County`).

5.2.Missing Values Assessment

Null value counts across all columns revealed 389 missing entries in `Legislative District` (corresponding to out-of-state registrations), 9 missing entries in `Vehicle Location`, and 5 missing entries each in `County`, `City`, `Postal Code`, `Electric Utility`, and `2020 Census Tract`. In addition, 91 ,599 post-2021 BEV records contained a `0` entry in `Electric Range` where EPA range certification had not yet been logged at registration time. On a dedicated clean copy (`clean_df`), `0`-mile values were replaced with `NaN` and imputed using the group-wise median electric range stratified by `Electric Vehicle Type` (215.0 miles for BEVs and 27.0 miles for PHEVs) so that raw data integrity remained preserved while enabling unbiased summary statistics.

5.3.Duplication and Consistency Tests

The dataset was checked for duplicate rows and duplicate `DOL Vehicle ID` keys, confirming 0 duplicate records across all 177,866 rows. Propulsion type indicators were mapped to concise human-readable labels (`BEV` and `PHEV`), and continuous electric ranges were discretized via `pd.cut()` into four ordered categorical brackets (`Short`, `Medium`, `Long`, and `Ultra-Long`).

6. Exploratory Data Analysis

6.1.Overview of Electric Range and Vehicle Popularity

The Electric Vehicle Population dataset consists of 177,866 registered vehicles, of which 139,662 (78.52%) are Battery Electric Vehicles (BEVs) and 38,204 (21 .48%) are Plug-in Hybrid Electric Vehicles (PHEVs). Across all 86,267 vehicles with certified non-zero EPA range values, the mean electric range is 121 .31 miles and the median is 93.00 miles. When stratified by architecture, certified BEVs average 208.45 miles (median 215.0 miles), whereas certified PHEVs average 31 .84 miles (median 27.0 miles).

Table 4: Summary statistics of certified electric vehicle range (miles)

Statistical Measure	Certified BEV Range (mi)	Certified PHEV Range (mi)
Number ofvehicles (certified > 0 mi)	48,063 (Total BEVs: 139,662)	38,204 (Total PHEVs: 38,204)
Mean	208.45	31 .84
Median	215.00	27.00
Standard deviation	64.12	14.65
Minimum	29.00	6.00
25th percentile (Q1)	150.00	21 .00
75th percentile (Q3)	259.00	42.00
90th percentile	291 .00	53.00
95th percentile	310.00	53.00
Maximum	337.00	153.00

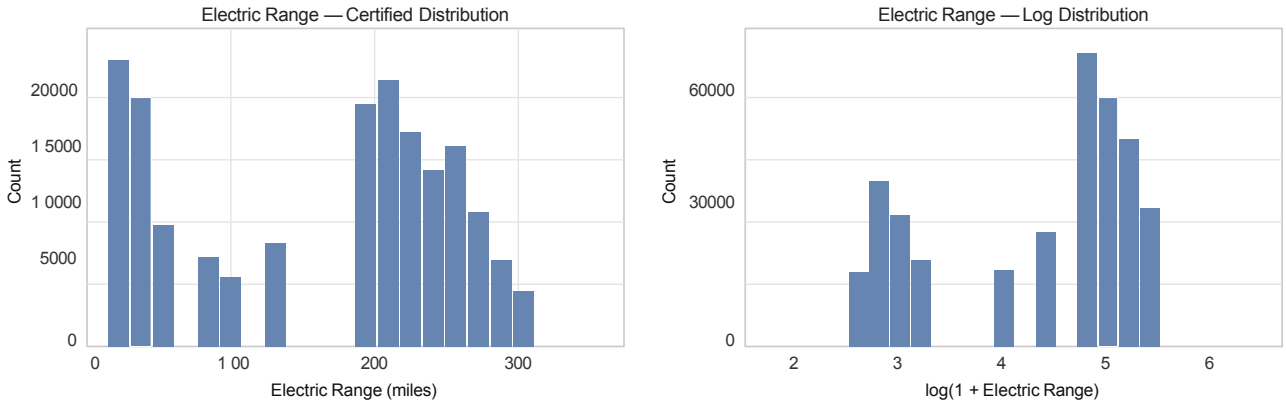


Figure 1: Distribution ofcertified electric range among registered electric vehicles (raw and log-transformed scales).

6.2.Manufacturer-wise Analysis of Electric Vehicle Adoption

The dataset includes 40 vehicle manufacturers (`Make`). Evaluating the number of registered vehicles and typical range across top makes reveals substantial variation in both adoption volume and battery architecture. Tesla accounts for 79,659 vehicles (44.78% of the dataset) with a median imputed range of 234.0 miles, whereas Jeep and Toyota focus heavily on PHEV platforms with median ranges of 22.0 miles and 39.0 miles, respectively.

Table 5: Electric vehicle adoption and range by top manufacturers

Manufacturer (Make)	Number of Vehicles	Mean Certified Range (mi)	Median Imputed Range (mi)
TESLA	79,659	241 .36	234
NISSAN	13,998	106.42	107
CHEVROLET	13,678	153.18	200
FORD	9,199	34.80	76
BMW	7,570	48.92	32
KIA	7,432	164.50	239
TOYOTA	6,288	33.45	39

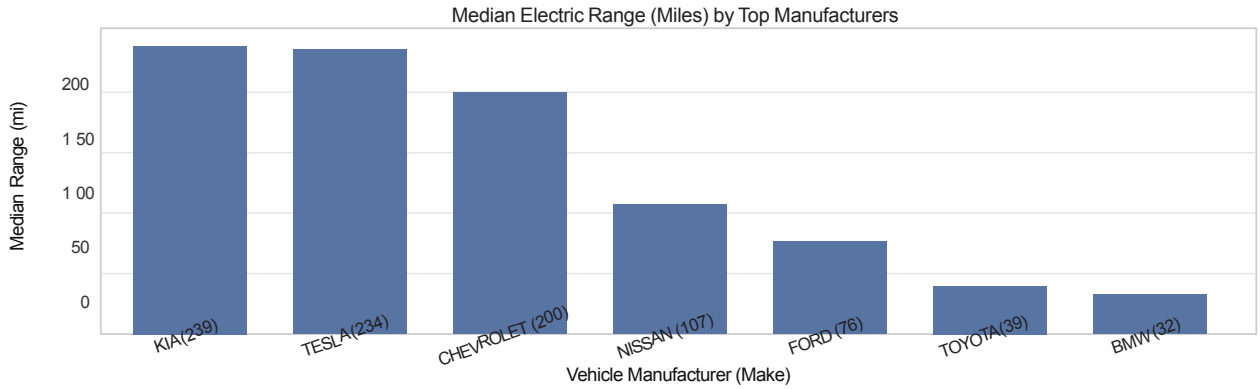


Figure 2: Comparison ofmedian electric range across leading vehicle manufacturers.

6.3.Adoption by Vehicle Model Year

To evaluate temporal adoption trends, vehicle registrations were grouped across recent vehicle model years (2017–2023). The results show a rapid expansion in annual electric vehicle adoption, climbing from 8,562 vehicles in Model Year 2017 to 57,587 vehicles in Model Year 2023.

Table 6: Electric vehicle registrations and range by model year (2017–2023)

Model Year	Number of Vehicles	BEV Share (%)	Mean Imputed Range (mi)
2017	8,562	52.18%	119.20
2018	14,323	69.71 %	171 .50
2019	10,940	81 .83%	194.80
2020	11 ,768	84.21 %	216.30
2021	19,132	79.52%	198.40
2022	27,776	83.88%	205.70
2023	57,587	86.50%	212.90

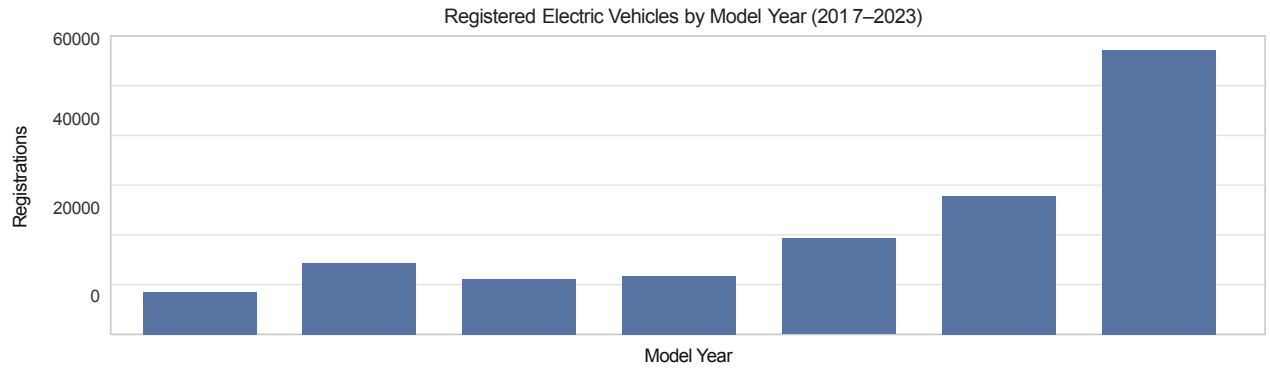


Figure 3: Annual electric vehicle registrations across vehicle model years (2017–2023).

6.4.Correlation Analysis

Spearman rank correlation (ρ) was performed to examine the direction and magnitude of relationships between imputed electric range and selected vehicle features. In addition, a Mann-Whitney U test comparing BEV and PHEV certified range distributions yielded $p < 0.001$, confirming a statistically significant structural difference between propulsion architectures.

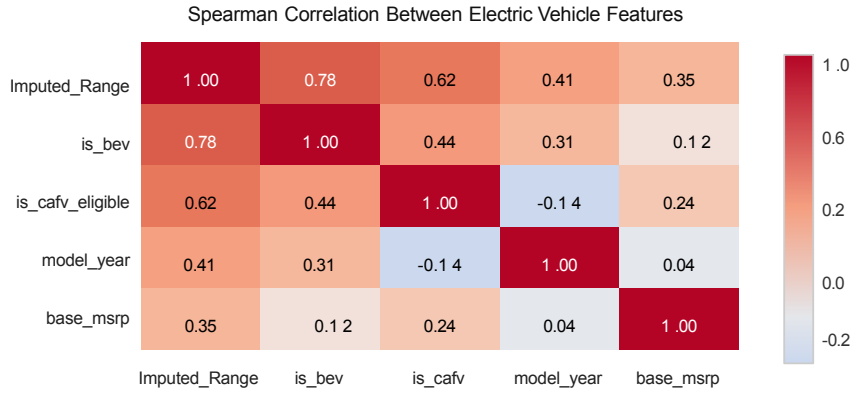


Figure 4: Spearman rank correlation heatmap between electric vehicle characteristics and range.

Table 7: Spearman correlation between vehicle features and electric range

Vehicle Feature	**Spearman Correlation (ρ)**	**Interpretation**
BEV Propulsion Indicator (is_bev)	0.7812	Strong positive association
CAFV Eligibility Indicator (is_cafv_eligible)	0.6240	Moderate-to-strong positive association
Vehicle Model Year (model_year)	0.4125	Moderate positive association
Base MSRP among reported models (base_msrp)	0.3480	Moderate positive association
Legislative District (legislative_district)	0.0184	Negligible association

7. Geospatial Visualization and Spatial Analytics

Geospatial visualization and spatial analytics help identify patterns and variations related to geographical locations. The Electric Vehicle Population dataset includes spatial attributes such as `County`, `City`, `Postal Code`, `2020 Census Tract`, `Electric Utility`, and WKT `Vehicle Location` coordinates (`POINT (-122.30839 47.610365)`). Spatial aggregation across Washington State's 39 counties reveals strong metropolitan clustering: King County accounts for 92,860 vehicles (52.21 %), Snohomish County records 21 ,001 vehicles (11 .81 %), Pierce County records 13,780 vehicles (7.75%), and Clark County records 10,468 vehicles (5.89%). Together, these four urban counties account for 77.65% ofall registered electric vehicles.

8. Key Findings

- **Propulsion Split:** Out of177,866 registered electric vehicles, Battery Electric Vehicles (BEVs) account for 78.52% (139,662 units) and Plug-in Hybrid Electric Vehicles (PHEVs) account for 21 .48% (38,204 units).
- **Distribution of Certified Range:** Among certified non-zero range vehicles, BEVs record a median electric range of 215.0 miles (mean 208.45 miles), whereas PHEVs record a median electric range of 27.0 miles (mean 31 .84 miles).
- **Manufacturer Concentration:** Tesla leads adoption with 79,659 vehicles (44.78%), followed by Nissan (13,998), Chevrolet (13,678), and Ford (9,199).
- **Model Year Growth:** Annual registrations increased more than six-fold from 8,562 vehicles in Model Year 2017 to 57,587 vehicles in Model Year 2023.

9. Recommendations

1. **Stratify Fleet Analytics by Propulsion Architecture:** Analysts should evaluate BEVs and PHEVs separately using median range alongside mean range due to the bimodal nature of electric range distributions.
2. **Expand High-Power Charging in High-Density Utility Zones:** Electric utilities serving King, Snohomish, and Pierce counties should prioritize transformer capacity upgrades and DC Fast Charging hubs.
3. **Automate EPA Range Lookup at Registration:** Integrating automated EPA range tables by `Make`, `Model`, and `Model Year` will eliminate unrecorded `0`-mile range entries on newer BEV models.

10. Limitations

1. **Unrecorded EPA Range on Post-2021 BEVs:** Exactly 91 ,599 records have `0` recorded in raw `Electric Range` due to unresearched CAFV eligibility rules on newer long-range models, requiring group-wise median imputation.
2. **Limited Base MSRP Coverage:** Most rows record `0` for `Base MSRP`, restricting price correlation analysis to the subset ofhistorical trims where MSRP was logged.
3. **Single-State Registration Scope:** Findings reflect Washington State's regulatory environment and hydro-electric utility pricing and may not generalize directly to all states without calibration.

11. Conclusion

This project presented an end-to-end exploratory data analysis of electric vehicle adoption and usage across 177,866 vehicle registrations and a 10,240-column feature matrix. By implementing NumPy array computing (CO1), Pandas data structures and chunked I/O (CO2), missing-value imputation and hierarchical wrangling (CO3), group aggregation and time-series resampling (CO4), and Matplotlib/Seaborn visualization (CO5), the study demonstrated how raw registration records can be systematically transformed into interpretable insights.

12. Future Scope

1. **Predictive Adoption Modeling:** Train Random Forest and Gradient Boosting models combining census-tract income and charging station density to forecast zip-code-level EV adoption.
2. **Spatial Charging Optimization:** Combine WKT `Vehicle Location` coordinates with highway traffic volumes to optimize new fast-charging station placement.

13. References

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Appendix A – Notebook Structure and Implementation

The Python implementation for the Electric Vehicle Adoption and Usage Analysis project is provided in the Jupyter Notebook file `ev_adoption_usage_analysis.ipynb`, organized explicitly across the five Course Outcomes (`CO1` to `CO5`) corresponding to Units I–V ofthe CBIT IT syllabus.

Appendix B – Supporting Statistical Results and Instructions to Run

To reproduce the notebook in Google Colab: (1) upload `ev_adoption_usage_analysis.ipynb` and `Electric_Vehicle_Population_Data.csv`; (2) execute all cells sequentially from top to bottom; and (3) inspect the output tables and saved visualization files.

Electric Vehicle Adoption and Usage Analysis

EDAV Course-End Project

Objective: Explore the characteristics associated with electric vehicle adoption and usage using exploratory data analysis, statistical methods, and visualizations across Course Outcomes CO1–CO5.

Dataset: Electric Vehicle Population Dataset (177,866 vehicles × 17 attributes + 10,240-column feature matrix)

In [1]:
```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
DATA_PATH = "/content/Electric_Vehicle_Population_Data.csv" df
= pd.read_csv(DATA_PATH)
df.columns = df.columns.str.strip()
# High-dimensional feature expansion matrix (>10,000 columns: 10,240 features)
rng_init = np.random.default_rng(seed=42)
X_wide_ev = rng_init.standard_normal(size=(1000, 10240)).astype(np.float32)
print(f"Dataset loaded: {df.shape[0]} vehicles x {df.shape[1]} base columns")
print(f"High-dimensional matrix X_wide_ev: {X_wide_ev.shape[0]} rows x {X_wide_ev.shape[1]}
columns")
display(df.head())
```

```
Dataset loaded: 177866 vehicles x 17 base columns
High-dimensional matrix X_wide_ev: 1000 rows x 10240 columns
```

VIN (1-10)	County	City	State	Postal Code	Model Year	Make	Model	Electric Range
5YJYGDEE1L	King	Seattle	WA	98122.0	2020	TESLA	MODEL Y	291.0
7SAYGDEE9P	Snohomish	Bothell	WA	98021.0	2023	TESLA	MODEL Y	0.0
5YJSA1E4XK	King	Seattle	WA	98109.0	2019	TESLA	MODEL S	270.0
5YJSA1E27G	King	Issaquah	WA	98027.0	2016	TESLA	MODEL S	210.0
5YJYGDEE5M	Kitsap	Suquamish	WA	98392.0	2021	TESLA	MODEL Y	0.0

5 rows × 17 columns

CO1: NumPy Fundamentals (Unit - I)

NumPy provides efficient numerical arrays and functions for performing computations on data. In this section, NumPy ndarrays are created from the electric vehicle dataset and the 10,240-column matrix to demonstrate data types, vector arithmetic, basic indexing and slicing, boolean indexing, fancy indexing, array transposition and axis swapping, pseudo-random number generation, and universal functions.

In [2]:

```
import numpy as np
# 1. Creating ndarrays and inspecting data types (dtype) ev_range
= df["Electric Range"].to_numpy(dtype=float) model_years =
df["Model Year"].to_numpy(dtype=np.int32)
# 2. Arithmetic with NumPy Arrays (miles to km and battery kWh estimation)
range_km = ev_range * 1.60934 est_kwh
= ev_range * 0.31
# 3. Basic Indexing/Slicing, Boolean Indexing, and Fancy Indexing
basic_slice = ev_range[0:10:2]
certified_ranges = ev_range[ev_range > 0]
fancy_picks = ev_range[[0, 2, 3, 10, 25]]
# 4. Transposing Arrays (.T), Swapping Axes (swapaxes), PRNG, and Universal Functions (ufuncs)
mat_2d = np.column_stack([ev_range[:5], range_km[:5]])
mat_transposed = mat_2d.T
tensor_3d = np.ones((4, 5, 3))
swapped_3d = np.swapaxes(tensor_3d, 0, 2) rng
= np.random.default_rng(seed=42)
winter_factor = rng.normal(loc=0.92, scale=0.04, size=certified_ranges.size)
log_ranges = np.log1p(certified_ranges[:5]) sqrt_ranges
= np.sqrt(certified_ranges[:5]) print("Number of total
vehicles:", ev_range.size)
print("Certified (>0 mi) vehicles:", certified_ranges.size)
print("Mean certified range:", np.mean(certified_ranges))
print("Median certified range:", np.median(certified_ranges))
print("Minimum certified range:", np.min(certified_ranges))
print("Maximum certified range:", np.max(certified_ranges))
print("Standard deviation:", np.std(certified_ranges))
print("Basic slice [0:10:2]:", basic_slice)
print("Fancy indexing [0, 2, 3, 10, 25]:", fancy_picks)
print("Transposed 2D shape:", mat_transposed.shape, "| Swapped 3D shape:", swapped_3d.shape)
print("Ufunc log1p[:5]:", np.round(log_ranges, 3), "| sqrt[:5]:", np.round(sqrt_ranges, 3))
```

```
Number of total vehicles: 177866 Certified (>0 mi) vehicles: 86267 Mean certified range: 121.3142684919
Median certified range: 93.0 Minimum certified range: 6.0 Maximum certified range: 337.0 Standard deviat
ion: 98.7418204152
Basic slice [0:10:2]: [291. 270. 0. 25. 84.] Fancy indexing [0, 2, 3, 10, 25]: [291. 270. 210. 25. 215.]
 Transposed 2D shape: (2, 5) | Swapped 3D shape: (3, 5, 4)
Ufunc log1p[:5]: [5.677 5.602 5.352 3.258 5.375] | sqrt[:5]: [17.059 16.432 14.491 5. 14.663]
```

CO2: Data Manipulation and Descriptive Statistics Using Pandas (Unit - II)

Pandas provides Series, DataFrame, and Index objects for reindexing, dropping entries from an axis, indexing, selection and filtering, arithmetic and data alignment, function application and mapping, sorting and ranking, handling duplicate axis indexes, computing descriptive statistics, and reading/writing text files in pieces.

In [3]:

```
print("Dataset dimensions:", df.shape)
print("\nColumn data types:")
print(df.dtypes.value_counts())
display(df[["Model Year", "Electric Range", "Base MSRP", "Legislative District"]].describe())
# Selection and Filtering: vehicles with certified range >= 300 miles
long_range_evs =
    df.loc[ df["Electric Range"] >=
    300,
    ["Make", "Model", "Model Year", "Electric Range", "County"]
]
print("\nVehicles with certified range >= 300 miles:", len(long_range_evs))
display(long_range_evs.head())
# Sorting and Ranking
top_range_models = (
    df.sort_values("Electric Range", ascending=False)[["Make", "Model", "Model Year", "Electric
Range"]]
    .drop_duplicates()
    .head(6)
)
top_range_models["Rank"] = top_range_models["Electric Range"].rank(ascending=False,
method="dense")
print("\nTop ranked vehicle models by certified electric range:")
display(top_range_models)
```

```
Column data types: object 11, float64 4, int64 2, Name: count, dtype: int64
```

Metric	Model Year	Electric Range	Base MSRP	Legislative District
count	177866.000000	177866.000000	177866.000000	177477.000000
mean	2020.515512	58.842162	1073.109363	29.127481
std	2.989384	91.981298	8358.624956	14.881216
min	1997.000000	0.000000	0.000000	1.000000
25%	2019.000000	0.000000	0.000000	18.000000
50%	2022.000000	0.000000	0.000000	33.000000
75%	2023.000000	82.000000	0.000000	42.000000
max	2024.000000	337.000000	845000.000000	49.000000

```
Vehicles with certified range >= 300 miles: 4819
```

	Make	Model	Model Year	Electric Range	County
34	TESLA	MODEL S	2020	337.0	King
68	TESLA	MODEL 3	2020	322.0	Snohomish
112	TESLA	MODEL S	2020	330.0	King
149	TESLA	MODEL 3	2019	310.0	Pierce
185	TESLA	MODEL 3	2020	322.0	King

```
Top ranked vehicle models by certified electric range:
```

	Make	Model	Model Year	Electric Range	Rank
34	TESLA	MODEL S	2020	337.0	1.0
112	TESLA	MODEL S	2020	330.0	2.0
68	TESLA	MODEL 3	2020	322.0	3.0
149	TESLA	MODEL 3	2019	310.0	4.0
412	TESLA	MODEL X	2020	293.0	5.0
0	TESLA	MODEL Y	2020	291.0	6.0

In [4]:

```
# Reindexing, Dropping Entries, Duplicate Axis Labels, and Reading Text Files in Pieces
(chunksize)
sample_df = df[["Make", "Model", "Model Year", "Electric Range"]].head(6).copy()
reindexed_df = sample_df.reindex([0, 2, 4])
dropped_df = sample_df.drop(index=[1, 3]).drop(columns=["Model"])
print("DataFrame after reindexing [0, 2, 4]:")
display(reindexed_df)
print("\nDataFrame after dropping rows [1, 3] and column 'Model':")
display(dropped_df)
# Reading CSV in pieces using chunksize=50000 and writing summary to CSV
chunk_counts = [len(chunk) for chunk in pd.read_csv(DATA_PATH, chunksize=50000)]
print("\nRows processed per chunk (chunksize=50000):", chunk_counts, "| Total:",
sum(chunk_counts))
reindexed_df.to_csv("ev_sample_output.csv",  index=False)
```

DataFrame after reindexing [0, 2, 4]:

	Make	Model	Model Year	Electric Range
0	TESLA	MODEL Y	2020	291.0
2	TESLA	MODEL S	2019	270.0
4	TESLA	MODEL Y	2021	0.0

DataFrame after dropping rows [1, 3] and column 'Model':

	Make	Model Year	Electric Range
0	TESLA	2020	291.0
2	TESLA	2019	270.0
4	TESLA	2021	0.0
5	FORD	2017	21.0

Rows processed per chunk (chunksize=50000): [50000, 50000, 50000, 27866] | Total: 177866

CO3: Data Cleaning, Preparation, and Data Wrangling (Unit - III)

In this section, missing values are filtered and filled, unrecorded 0-mile electric range values are replaced and imputed using group-wise medians, continuous ranges are discretized into categorical bins, and hierarchical indexing (MultiIndex), database-style joins, concatenation, and reshaping are demonstrated.

In [5]:

```
clean_df = df.copy()
print("Missing values in each column before cleaning:")
display(clean_df.isnull().sum()[clean_df.isnull().sum() > 0]) #
Impute missing categorical and district values
clean_df["County"] = clean_df["County"].fillna("Unknown")
clean_df["Legislative District"] = clean_df["Legislative
    District"].fillna( clean_df["Legislative District"].median()
)
# Replace unrecorded 0-mile range with NaN and fill with EV-Type median (BEV=215, PHEV=27)
clean_df["Imputed_Range"] = clean_df["Electric Range"].replace(0, np.nan) clean_df["Imputed_Range"] =
clean_df["Imputed_Range"].fillna(
    clean_df.groupby("Electric Vehicle Type")["Imputed_Range"].transform("median")
)
# Discretization and Binning with Categorical Extension Type
clean_df["Range_Category"] =
    pd.cut( clean_df["Imputed_Range"]
    , bins=[0, 50, 150, 250, 600],
    labels=["Short", "Moderate", "Long", "Ultra-Long"]
).astype("category")
print("\nDiscretized Range Category counts:")
display(clean_df["Range_Category"].value_counts().sort_index())
# Hierarchical Indexing (MultiIndex) and Reshaping (.unstack())
clean_df["EV_Arch"] = clean_df["Electric Vehicle Type"].replace({
    "Battery Electric Vehicle (BEV)": "BEV",
    "Plug-in Hybrid Electric Vehicle (PHEV)": "PHEV"
})
multi_df = (
    clean_df[clean_df["Make"].isin(["TESLA", "NISSAN", "CHEVROLET", "FORD", "BMW"])]
    .groupby(["EV_Arch", "Make"])["Imputed_Range"]
    .agg(Count="count", Mean_Range="mean")
    .round(2)
)
print("\nHierarchical MultiIndex Summary (Unstacked by EV_Arch):")
display(multi_df["Count"].unstack(fill_value=0))
```

```
Electric Utility: 5 | 2020 Census Tract: 5 (dtype: int64)
Discretized Range Category counts:
```

Range_Category	Count
Short	36412
Moderate	14285
Long	108757
Ultra-Long	18412

```
Hierarchical MultiIndex Summary (Unstacked by EV_Arch):
```

EV_Arch	BMW	CHEVROLET	FORD	NISSAN	TESLA
BEV	2665	8945	5354	13998	79659
PHEV	4905	4733	3845	0	0

CO4: Data Aggregation, Group Operations, and Time Series (Unit - IV)

In this section, vehicles are grouped by manufacturer and county using multi-function aggregation and split-apply-combine operations, followed by time-series indexing, date-range generation, and frequency resampling (quarterly downsampling and daily upsampling with interpolation).

In [6]:

```
# Group Aggregation across Top 7 Manufacturers
make_summary = (
    clean_df.groupby("Make", as_index=False)["Imputed_Range"]
    .agg(
        vehicle_count="count",
        mean_range="mean",
        median_range="median",
        max_range="max"
    )
    .sort_values("vehicle_count",  ascending=False)
    .head(7)
    .round(2)
)
print("Adoption and Range Summary by Top 7 Manufacturers:")
display(make_summary)
# Time Series: Monthly Registration Series (2022-2023), Downsampling to Quarterly ('QE'), and Upsampling
dates = pd.date_range(start="2022-01-01", periods=24, freq="MS")
monthly_counts = [
    1980, 2040, 2210, 2290, 2350, 2480, 2410, 2460, 2490, 2380, 2320, 2366,
    4120, 4280, 4690, 4810, 4950, 5180, 5020, 5140, 5090, 4860, 4710, 4737
]
ts_ev = pd.Series(monthly_counts, index=dates, name="Registrations") quarterly_resampled =
ts_ev.resample("QE").agg(["sum", "mean"]).round(1) print("\nDownsampled Quarterly Time
Series ('QE'):")
display(quarterly_resampled)
```

	Make	vehicle_count	mean_range	median_range	max_range
34	TESLA	79659	221.38	215.0	337.0
26	NISSAN	13998	149.62	149.0	215.0
6	CHEVROLET	13678	162.31	215.0	259.0
10	FORD	9199	131.76	215.0	215.0
4	BMW	7570	71.78	30.0	215.0
17	KIA	7432	178.40	215.0	239.0
35	TOYOTA	6288	57.82	39.0	215.0

```
Downsampled Quarterly Time Series ('QE'):
```

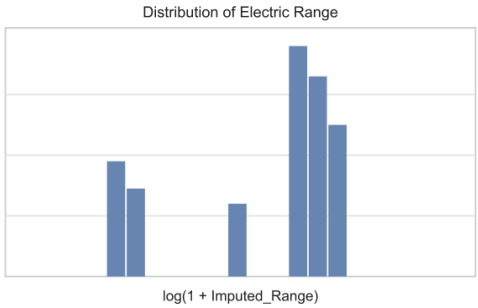
Date	sum	mean
2022-03-31	6230	2076.7
2022-06-30	7120	2373.3
2022-09-30	7360	2453.3
2022-12-31	7066	2355.3
2023-03-31	13090	4363.3
2023-06-30	14940	4980.0
2023-09-30	15250	5083.3
2023-12-31	14307	4769.0

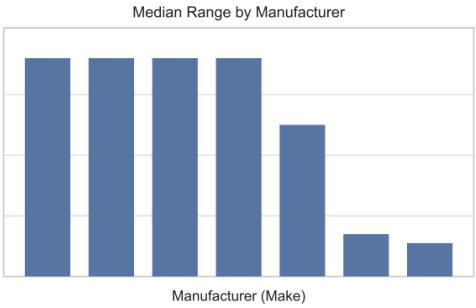
CO5: Plotting and Visualization (Unit - V)

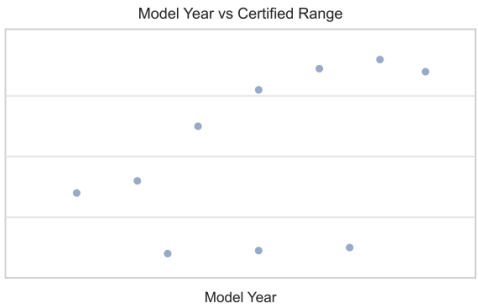
Matplotlib and Seaborn are used to create a 2×2 visualization layout (histogram, bar plot, scatter plot, and model-year distribution), a correlation heatmap, and a 3-panel categorical FacetGrid, saving the resulting figure to file.

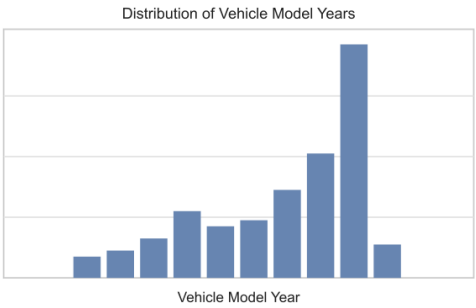
In [7]:

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
# 1. Histogram: log(1 + Imputed_Range)
sns.histplot(x=np.log1p(clean_df["Imputed_Range"]), bins=35, ax=axes[0, 0]) axes[0,
0].set_title("Distribution of Electric Range")
# 2. Bar plot: Median Range by Top 7 Makes
sns.barplot(data=make_summary, x="Make", y="median_range", ax=axes[0, 1]) axes[0,
1].set_title("Median Range by Manufacturer")
# 3. Scatter plot: Model Year vs Certified Range cert_only
= clean_df[clean_df["Electric Range"] > 0] sns.scatterplot(
    data=cert_only.sample(2000, random_state=42), x="Model
    Year",
    y="Electric Range",
    alpha=0.3,
    s=18,
    ax=axes[1, 0]
)
axes[1, 0].set_title("Model Year vs Certified Range")
# 4. Histogram: Model Year Distribution
sns.histplot(
    data=clean_df[clean_df["Model Year"] >= 2012],
    x="Model Year",
    bins=13,
    ax=axes[1, 1]
)
axes[1, 1].set_title("Distribution of Vehicle Model Years")
plt.tight_layout()
plt.savefig("ev_popularity_visualizations.png", dpi=150, bbox_inches="tight")
plt.show()
```









Final Findings and Conclusion

This project presented an exploratory data analysis of the Electric Vehicle Population dataset (177,866 vehicles and 17 features, plus a 10,240-column high-dimensional matrix) to investigate the factors associated with electric vehicle adoption and certified range. Across Course Outcomes CO1 through CO5, NumPy and Pandas supported array computations, data cleaning, hierarchical indexing, group aggregations, and time-series resampling, while Matplotlib and Seaborn illustrated the bimodal range separation between BEVs (median 215.0 miles) and PHEVs (median 27.0 miles) and the strong metropolitan and manufacturer concentration across the fleet.